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Overview

Title

Design and Analysis of Digital Control Systems ☆

Number

0000002158

Persons involved**Lecturer (Assistant)**

👤 [Cao, Hongpeng](#) 👤 [Trumpp, Raphael](#) 👤 [Caccamo, Marco](#)

Type

lecture (VO)

Semester weekly hours

2

ECTS credits

-

Course language/s

English

Offered in

Winter semester 2024/25

Organisation

[Chair of Cyber-Physical Systems in Production Engineering \(Prof. Caccamo\)](#)

Description

Content

This course provides an introduction to C programming, embedded systems, and control engineering.

The course is structured along these three main topics:

1. C Programming and coding structure
 - a. Programming languages, compilation and compilers
 - b. Errors and debugging
 - c. C syntax, variables, data types and operators
 - d. Flow control statements
 - e. Pointers and arrays
 - f. Type conversion and casting
 - g. Lifetime and scope of variables
 - h. Input and output libraries, file handling
2. Embedded systems for digital control systems: Arduino board
 - a. Embedded systems introduction
 - b. Designing programs for the Arduino platform
 - c. Digital and analog inputs and outputs

3. Introduction to control engineering

- a. System description and modeling
- b. Mathematical background: differential equations, Laplace transform
- c. Modeling and simulation with MATLAB/Simulink
- d. Proportional, integral and derivative control
- e. System stability and controller performance

Previous knowledge expected

The basic mathematics module of the first year for TUM-BWL students is a prerequisite. It is helpful to attend IN8005 Introduction into Computer Science (for non-Informatics studies, TUM BWL).

Objective

Upon successful completion of the module, the participants understand important concepts and methods required to model and analyze modern digital control systems. The participants are able to write simple programs in C independently and have an understanding of data types and program structure. They understand the importance of embedded systems for digital control systems and can develop software interacting with the physical environment. The students can apply the basic control engineering methods and simulate and analyze simple systems in MATLAB/Simulink.

Teaching and learning method

The module consists of a lecture and tutorial classes. The lecture will introduce new concepts, illustrated and supported with slides and live programming examples. This approach helps to impart basic knowledge and allows students to understand different methods to model and analyze modern digital control systems. Students will apply and deepen their newly acquired knowledge through 4-6 project-oriented assignments in groups to demonstrate the practical relevance of the concepts introduced and explained during the lecture. Solving the assignments is supported by the tutorial classes, allowing students to interact with experienced teaching staff directly.

Course Criteria and Registration

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Further information

Online information

[e-learning course \(moodle\)](#)

▼ Categories

Dates and Groups



^ Standardgruppe ☆

Lecturer [Caccamo, Marco](#) [Cao, Hongpeng](#) [Trumpp, Raphael](#)



Tuesday , 10:00 - 11:30 from 15.10.2024 to 04.02.2025

[Online: Videokonferenz / Zoom etc.](#)

Series

Support

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Status within Curriculum

^ My Studies

Management and Technology (Main location: Heilbronn) - Bachelor of Science (1630 17 247)
, 20201 (offered until 30.09.2050)

■ [20201] Bachelorstudiengang Management and Technology (Heilbronn)

Specialization in Technology

Specialization in Technology: Digital Technologies

Required Modules: Digital Technologies

★ [MW2383] Design and Analysis of Digital Control Systems

▲ Design and Analysis of Digital Control Systems

Subject type Compulsory elective subject | **Recommended** 3rd semester | **Preconditions** None

→ [Go to my programme](#)

▼ Other Degree Programmes

Equivalent courses

^ Winter semester 2024/25

0000002158

Design and Analysis of Digital Control Systems

VO | 2 SWS | - ECTS | 2024 W

Lecturer

👤 [Cao, Hongpeng](#)

👤 [Trumpp, Raphael](#)

👤 [Caccamo, Marco](#)

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